

EXPT NO:8	FORECASTING AND PREDICTIVE ANALYTICS USING TABLEAU AND POWER BI
DATE: 14.08.2026	

PRE-LAB QUESTIONS (PROVIDE BRIEF ANSWERS TO THE FOLLOWING QUESTIONS)

1. What is forecasting in the context of Business Intelligence (BI)?

Forecasting in Business Intelligence is the process of using historical and current data to predict future values and trends. It uses statistical and analytical techniques to estimate future sales, revenue, demand, or other business activities. Forecasting helps organizations make better decisions, plan resources, manage inventory, and set future targets.

2. Explain the difference between predictive analytics and descriptive analytics.

Descriptive analytics focuses on understanding what happened in the past by analyzing historical data and presenting it through reports, charts, and dashboards. Predictive analytics uses historical data, statistical techniques, and forecasting models to predict what is likely to happen in the future. For example, a sales report is descriptive analytics, while predicting next month's sales is predictive analytics.

3. Name two common forecasting methods used in Tableau and Power BI.

Two common forecasting methods are:

- Exponential Smoothing – It gives more importance to recent observations while forecasting future values.
- Moving Average – It calculates the average of previous observations to identify trends and smooth out fluctuations in the data.

4. Why is data granularity important for accurate forecasting?

Data granularity refers to the level of detail at which data is stored or analyzed, such as daily, weekly, or monthly sales. Choosing the correct granularity helps identify important trends, patterns, and seasonal variations. If the data is too detailed or too general, important patterns may be missed. Therefore, proper granularity improves the accuracy and reliability of forecasting.

5. Name two challenges in forecasting using multiple data sources.

Two major challenges are:

- Inconsistent data formats: Different sources may have different date formats, column names, or data types.
- Missing or duplicate data: Missing values and duplicate records can affect the accuracy of the forecasting model.

IN-LAB EXERCISE:

OBJECTIVE:

To perform time series forecasting and predictive analytics on historical sales data using Tableau and Power BI, and visualize future trends for informed decision-making.

PROCEDURE:

1. Data Collection:

- Import historical sales data from multiple sources:
 - Dataset 1: Sales Data (CSV) → Order ID, Order Date, Region, Sales Amount
 - Dataset 2: Promotions Data (Excel) → Promotion ID, Region, Promotion Start Date, Promotion End Date, Discount

2. Data Preparation and Transformation:

- Clean data: standardize date formats, remove duplicates, handle missing values.
- Aggregate sales data at the required granularity (daily, weekly, or monthly).
- Ensure numeric fields (Sales, Discount) are correct.

3. Data Integration and Modeling:

- **Tableau:** Connect datasets and relate them using the Region field.
- **Power BI:** Create relationships in the Model view; define cardinality where necessary.

4. Forecasting and Predictive Analytics:

- **Tableau:**
 - Use the Forecast feature on Sales over time.
 - Apply trend models like Exponential Smoothing and visualize prediction intervals.
- **Power BI:**
 - Use built-in forecasting in line charts.
 - Optionally, create a DAX measure for predicted sales using moving averages or linear regression.

5. Create Calculated Fields:

- Forecasted Sales = Predicted value from the model
- Sales Growth % = $((\text{Forecasted Sales} - \text{Actual Sales}) / \text{Actual Sales}) * 100$

6. Design a Comprehensive Dashboard:

- Line chart showing actual vs. forecasted sales over time.
- KPI cards: Total Forecasted Sales, Expected Growth %.
- Use combo charts to show sales trends along with promotion periods.
- Highlight regions with expected growth above or below target thresholds.

7. Validate Forecast Accuracy:

- Compare forecasts against actual historical sales (back-testing).
- Adjust forecasting parameters if necessary (e.g., seasonality length, smoothing factor).

8. Publishing and Sharing:

- **Tableau:** Publish workbook to Tableau Public or Tableau Server.
- **Power BI:** Publish report to Power BI Service and share with stakeholders.

Experiment Datasets:

- **Dataset 1:** Historical Sales Data (CSV) → Order ID, Order Date, Region, Sales
- **Dataset 2:** Promotions Data (Excel) → Promotion ID, Region, Promotion Start/End Date, Discount

TABLEAU STEPS :

Tableau - Book1

File Data Server Window Help

sales_data.csv+ (Multiple Connections)

Connection: Live (selected), Extract

Filters: 0 | Add

sales_data.csv Promotions

Promotions 5 fields 28 rows

Name: Promotions

Description: No description available.

Fields:

Type	Field Name	Physical Table	Remot...
ABC	Promotion ID	Promotions	Promoti...
ABC	Region (Promotions)	Promotions	Region (...)
Calendar	Promotion Start Date	Promotions	Promoti...
Calendar	Promotion End Date	Promotions	Promoti...
Measure	Discount (%)	Promotions	Discount...

Promotion ID	Region (Promotions)	Promotion Start Date	Promotion End Date	Promotions Discount (%)
PROMO-5000	North	10-05-2024	20-05-2024	20
PROMO-5001	North	10-11-2024	20-11-2024	15
PROMO-5002	North	12-04-2023	22-04-2023	20
PROMO-5003	North	29-03-2022	03-04-2022	25
PROMO-5004	North	21-05-2024	04-06-2024	20
PROMO-5005	North	09-02-2024	23-02-2024	25
PROMO-5006	North	16-03-2024	23-03-2024	25
PROMO-5007	North	04-10-2022	09-10-2022	25
PROMO-5008	South	22-01-2022	27-01-2022	25
PROMO-5009	South	06-05-2024	11-05-2024	20
PROMO-5010	South	26-05-2023	02-06-2023	25
PROMO-5011	South	01-12-2023	15-12-2023	20
PROMO-5012	South	11-04-2024	16-04-2024	20
PROMO-5013	South	09-10-2023	14-10-2023	10

Tableau - Book1

File Data Server Window Help

sales_data.csv+ (Multiple Connections)

Connection: Live (selected), Extract

Filters: 0 | Add

sales_data.csv Promotions

sales_data.csv 4 fields 4180 rows

Name: sales_data.csv

Description: No description available.

Fields:

Type	Field Name	Physical Table	Remot...
ABC	Order ID	sales_data.csv	Order ID
Calendar	Order Date	sales_data.csv	Order Da...
ABC	Region	sales_data.csv	Region
Measure	Sales Amount	sales_data.csv	Sales A...

Order ID	Order Date	Region	Sales Amount
ORD-103189	02-04-2022	West	667.00
ORD-100944	30-09-2024	North	1270.49
ORD-100316	01-12-2022	North	755.29
ORD-102898	02-06-2024	East	1606.28
ORD-104083	04-11-2024	West	861.82
ORD-101332	09-11-2022	South	711.22
ORD-101243	06-08-2022	South	1025.23
ORD-102415	07-01-2023	East	444.79
ORD-101806	04-04-2024	South	886.17
ORD-103079	06-12-2024	East	1008.40
ORD-101612	08-09-2023	South	882.54
ORD-103022	08-10-2024	East	1246.00
ORD-103718	19-10-2023	West	316.48
ORD-100406	10-03-2023	North	975.33

Tableau - Book1

File Data Server Window Help

sales_data.csv+ (Multiple Connections)

Connection: ☒ Live ☐ Extract Filters: 0 | Add

sales_data (Text file) | promotions_data (Microsoft Excel)

Sheets: Promotions

sales_data.csv — Promotions

100 rows

Name: sales_data.csv

How do relationships differ from joins? [Learn more](#)

sales_data.csv Operator Promotions

Abc Region = Abc Region (Promoti

Add more fields

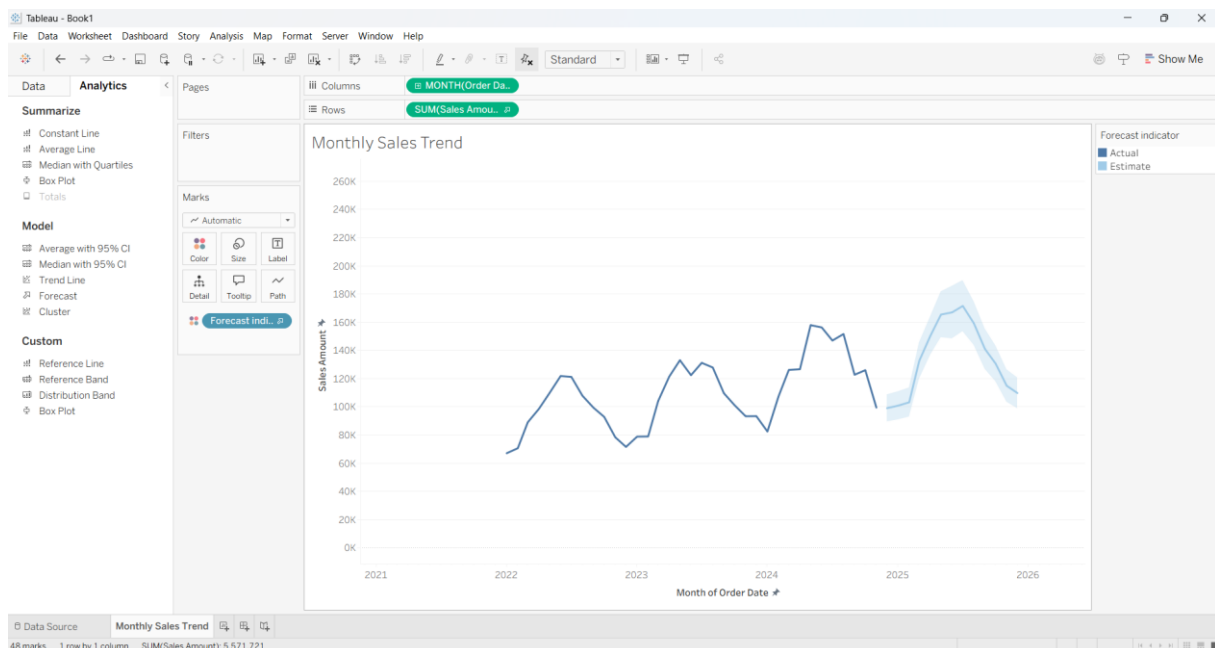
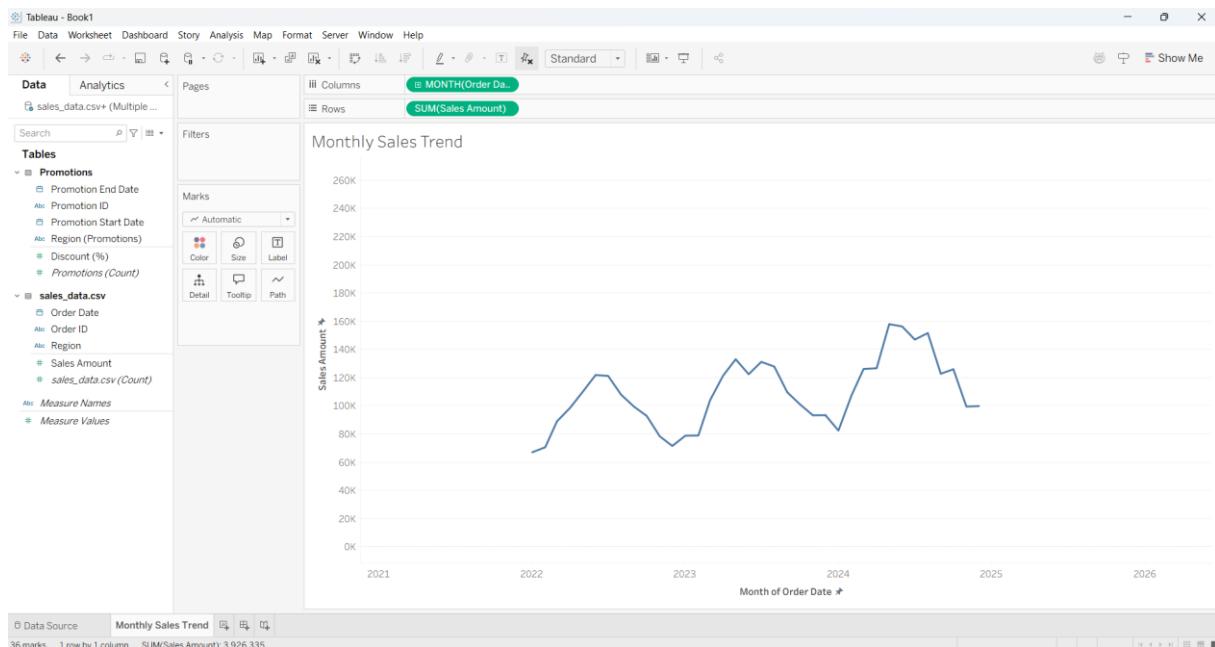
Performance Options

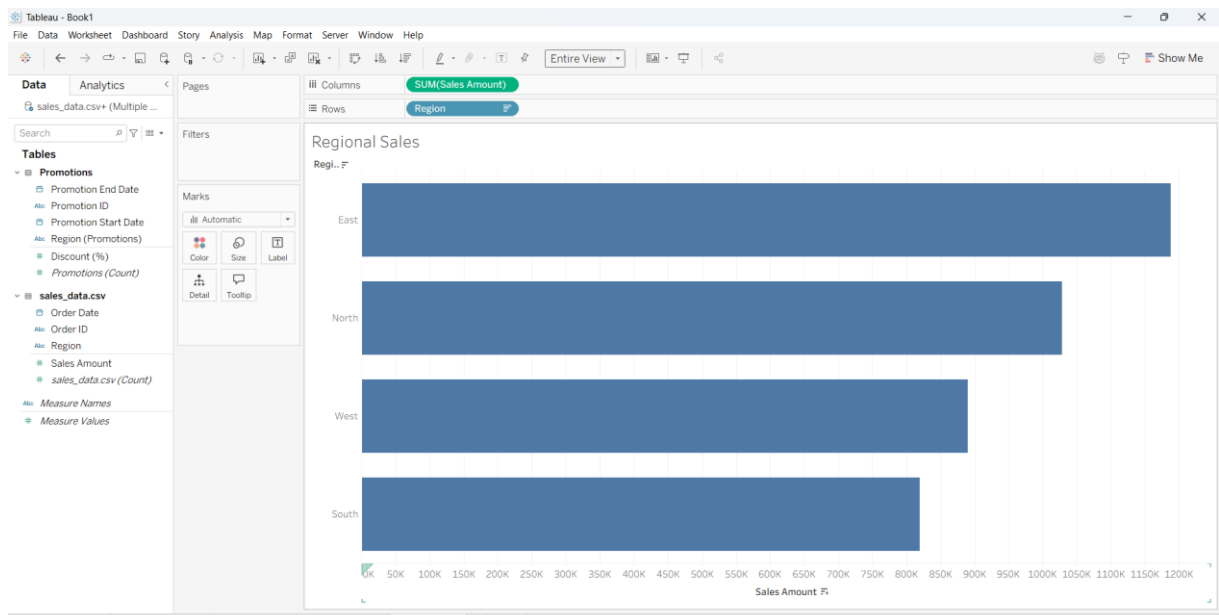
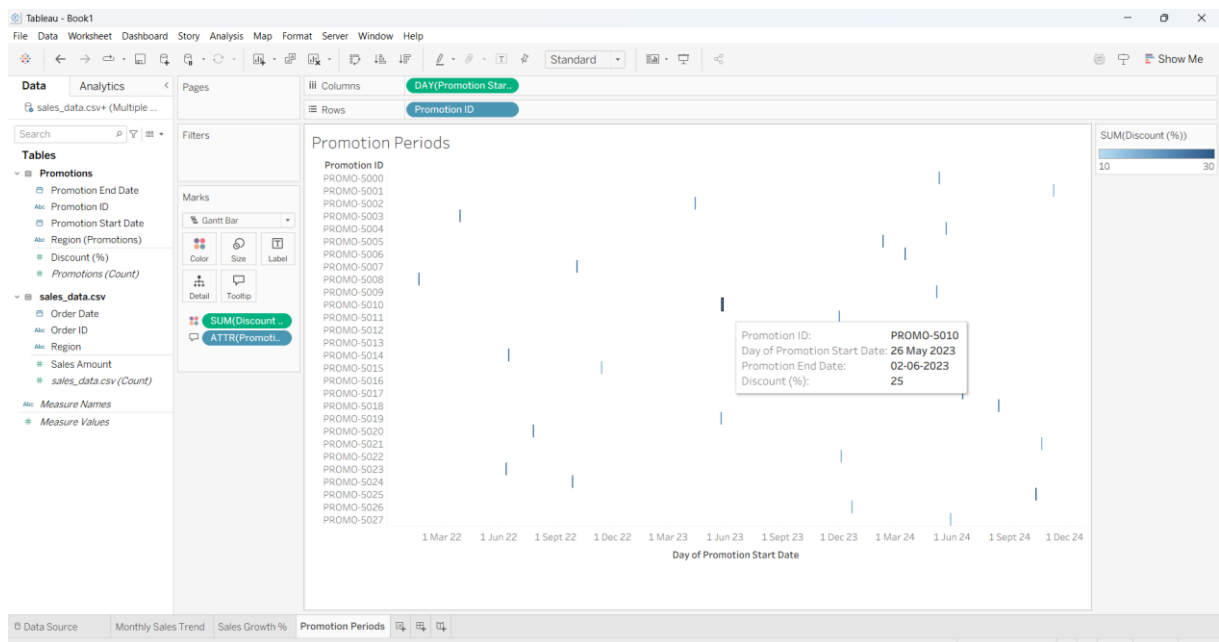
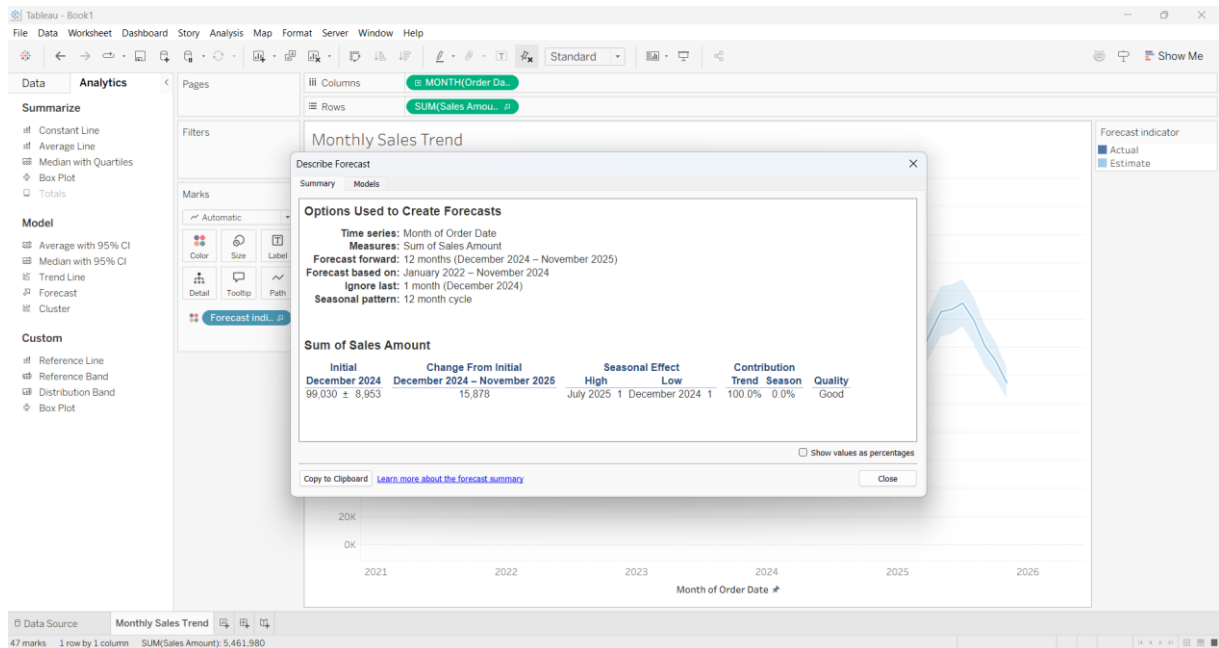
Remove Relationship

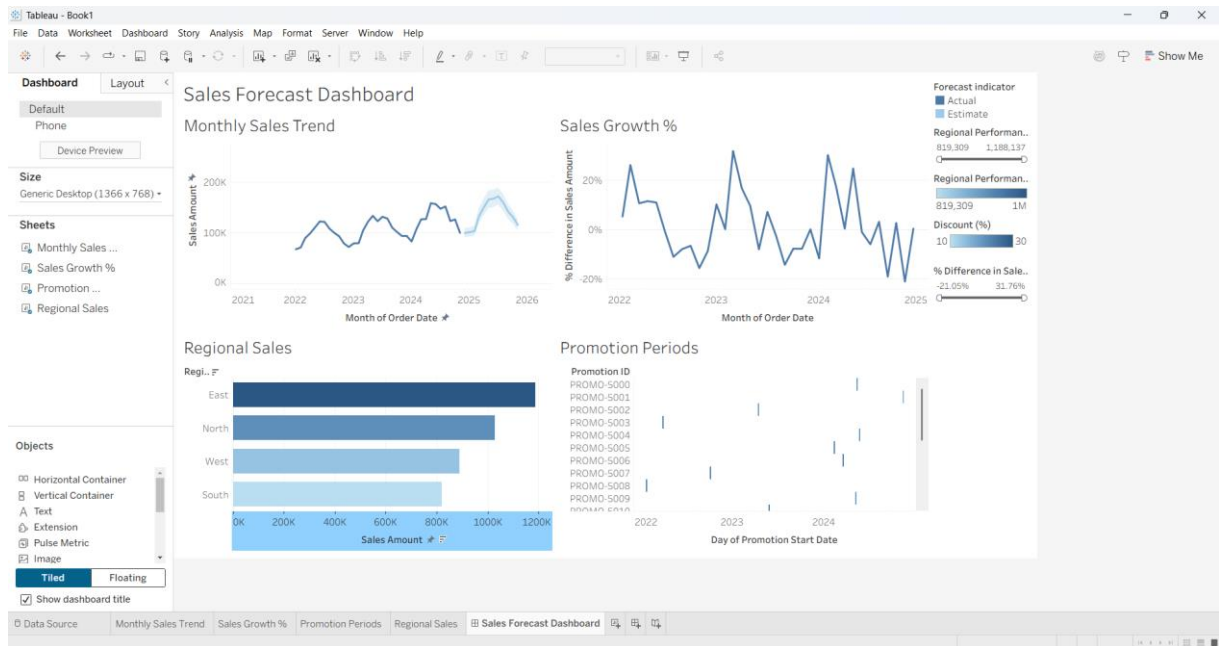
Order ID	Order Date	Region	Sales Amount
ORD-103189	02-04-2022	West	667.00
ORD-100944	30-09-2024	North	1270.49
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ORD-103022	08-10-2024	East	1246.00
ORD-103718	19-10-2023	West	316.48
ORD-100406	10-03-2023	North	975.33

Go to Worksheet

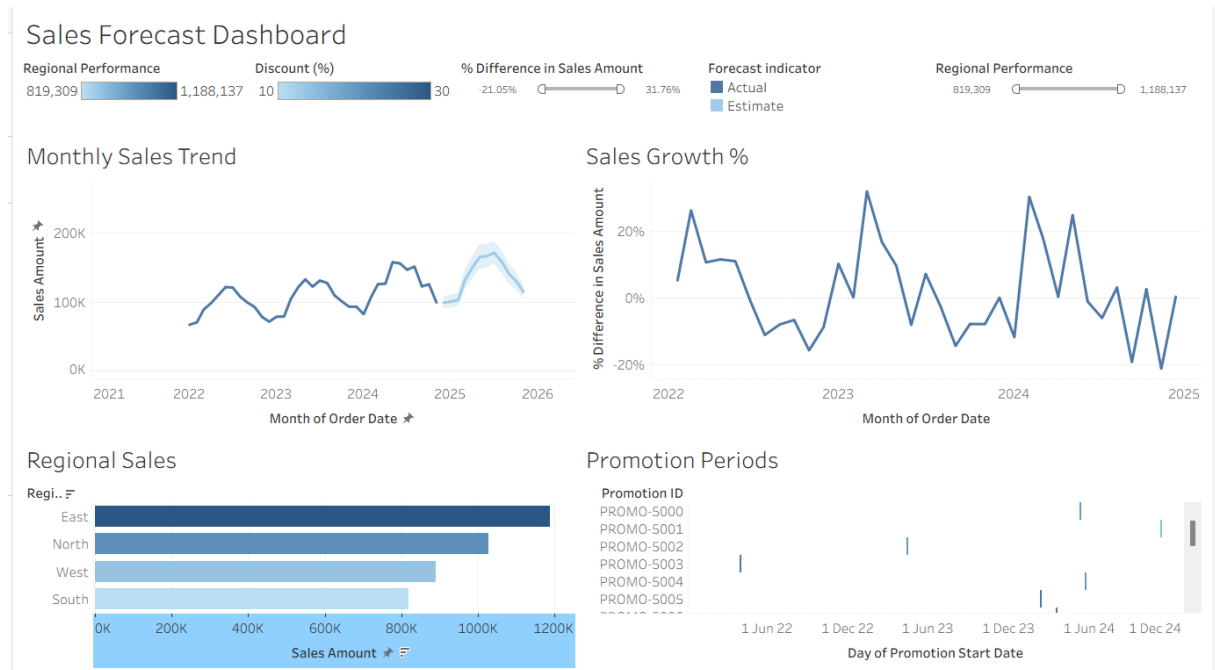
Data Source Sheet 1





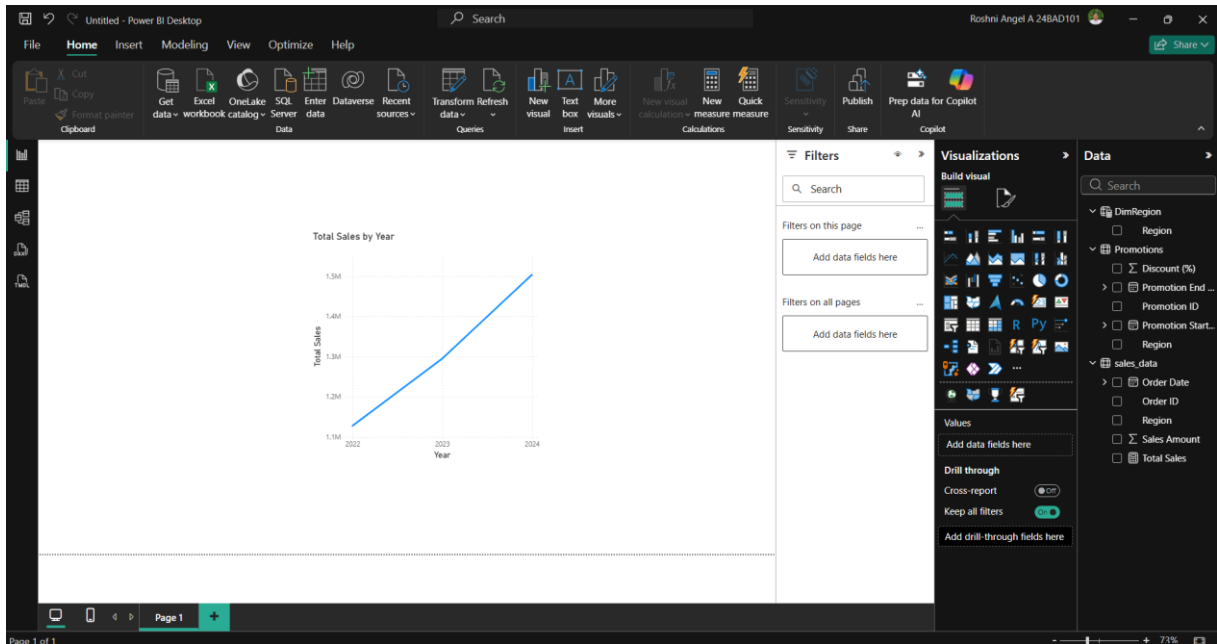


FINAL DASHBOARD :

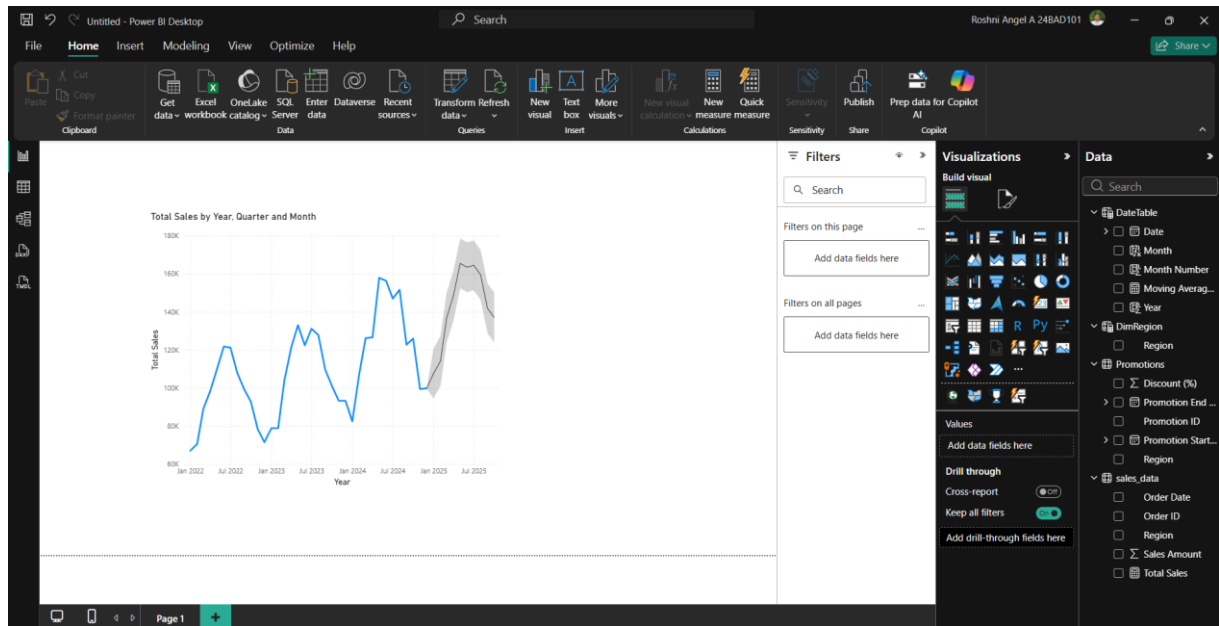


POWER BI STEPS :

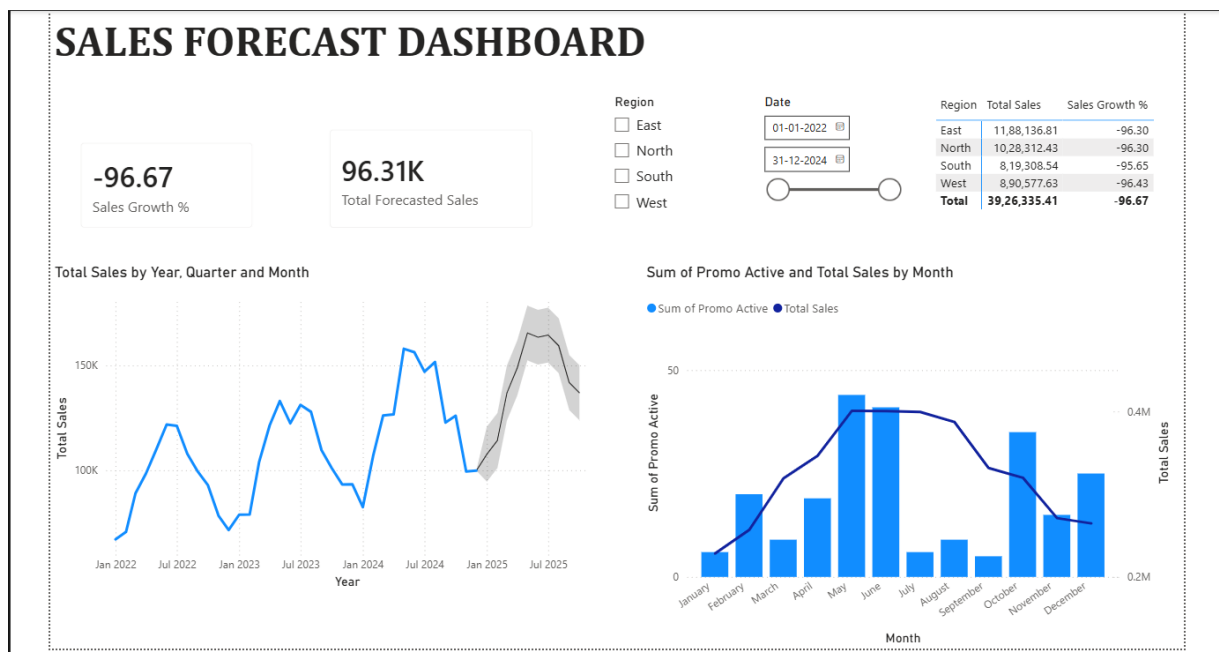
This screenshot shows the Power BI Desktop interface with the Data view selected. The main workspace displays three tables: **sales_data**, **Promotions**, and **DimRegion**. The **sales_data** table includes fields: Order Date, Order ID, Region, and Sales Amount. The **Promotions** table includes: Discount (%), Promotion End Date, Promotion ID, and Promotion Start Date. The **DimRegion** table includes: Region. Relationships are indicated by lines connecting the tables: a one-to-many relationship between **DimRegion** and **sales_data** on the 'Region' field, and a one-to-many relationship between **Promotions** and **sales_data** on the 'Promotion ID' field. The right-hand pane shows the 'Properties' tab with 'Cards' selected, and the 'Data' pane on the far right lists the available tables.



This screenshot shows the Power BI Desktop interface with the Relationship view selected. The main workspace displays the existing relationships between **sales_data**, **Promotions**, and **DimRegion**. A new relationship is being added between **sales_data** and **DateTable**. The 'Properties' pane on the right shows the relationship details: **Table** is **sales_data**, **Column** is **Order Date**, **Cardinality** is **Many to one (*:1)**, **Make this relationship active** is checked, **Cross-filter direction** is **Single**, and **Apply security filter in both directions** is unchecked. The 'Data' pane on the far right lists the available tables and fields.



FINAL DASHBOARD :



POWER BI DASHBOARD LINK :

https://kumaragurudtsteam-my.sharepoint.com/:u:/r/personal/roshniangel_24ad_kct_ac_in/Documents/Business%20intelligence/SALES%20FORECAST%20DASHBOARD.pbix?d=wd5e805cfa7f449ec87975f84ae072e1a&csf=1&web=1&e=xQ5CJI

POST-LAB QUESTIONS (PROVIDE BRIEF ANSWERS TO THE FOLLOWING QUESTIONS)

1. How did forecasting help in predicting future sales trends for business planning?

- Forecasting helped to estimate future sales by analyzing historical sales patterns and trends. It provided an expected range of future sales and helped identify whether sales were likely to increase or decrease.
- This information can help managers plan inventory, allocate resources, set sales targets, and organize promotional activities. Therefore, forecasting supports better and more data-driven business planning.

2. What forecasting method did you use, and why was it suitable for the dataset?

- I used Exponential Smoothing for forecasting the sales data. It was suitable because the dataset contains sales values recorded over time, making it a time-series dataset. Exponential Smoothing gives more importance to recent observations and can identify trends and seasonal patterns.
- Therefore, it provides a useful estimate of future sales based on previous sales behavior.

3. What challenges, if any, did you face regarding missing data or irregular time intervals?

- The main challenges were missing values, inconsistent date formats, and irregular time intervals in the datasets. These issues can affect the accuracy of a forecasting model.
- To handle them, the data was cleaned, duplicate records were removed, date formats were standardized, and the sales data was aggregated into a consistent time interval such as monthly data. This helped make the dataset suitable for forecasting.

4. How can predictive analytics influence decision-making in a retail environment?

- Predictive analytics helps retailers make better decisions by predicting future sales, customer demand, and product performance.
- It can help determine which products may have higher demand and when additional inventory is required.
- Retailers can also use predictions to plan discounts, promotions, pricing, and staffing. This reduces business risk and helps improve customer satisfaction and profitability.

5. What additional data could be integrated to improve forecast accuracy?

- Additional information such as competitor pricing, customer demand, discounts, promotional campaigns, holidays, weather conditions, economic indicators, and customer purchasing behavior can be integrated with the sales data.
- Competitor prices can help understand market changes, while holiday and promotion data can identify seasonal increases in sales. Combining these sources can provide more useful information and improve the accuracy of future sales forecasts.

RESULT:

The historical sales and promotions datasets were successfully cleaned, integrated, and modeled in Tableau and Power BI. Forecasting techniques provided predicted sales trends, growth percentages, and region-wise performance expectations. The resulting dashboards allowed managers to visualize actual vs. forecasted sales, enabling better planning and data-driven decision-making.

ASSESSMENT

Description	Max Marks	Marks Awarded
Pre Lab Exercise	5	
In Lab Exercise	10	
Post Lab Exercise	5	
Viva	10	
Total	30	
Faculty Signature		